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Dissertation

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Table of contents

5	2 CHAPTER 0	5
6	2.1 Motivation and Background	5
7	2.2 Overview of Physical 3D Visualizations	7
8	2.2.1 Creating 3D-Printed Visualizations	7
9	2.2.2 Current use cases	9
10	2.3 Bar Charts in Research and Practice	9
11	2.3.1 Against 3D Bar Charts	10
12	2.3.2 For 3D Bar Charts	11
13	2.3.3 Conflicting Empirical Studies	11
14	2.3.4 Construction of 3D Bar Charts	18
15	2.4 Heat Maps in Research and Practice	18
16	2.4.1 Empirical Studies of 2D and 3D Heat Maps	20
17	2.5 Methods for Evaluating Visualization Effectiveness	20
18	2.5.1 What makes a good chart?	20
19	2.5.2 Designing studies for numerical estimations	22
20	3 CHAPTER 2	25
21	3.1 Abstract	25
22	3.2 Introduction	26
23	3.3 Methods	28
24	3.3.1 Overview and Motivation	28
25	3.3.2 Experiential Learning	28
26	3.3.3 Data Analysis	30
27	3.3.4 Graphics Experiment	31
28	3.4 Results	33
29	3.4.1 Recruitment of Students and Instructors	33
30	3.4.2 Selected Responses from Experiment Participation	33
31	3.4.3 Selected Responses from Research Reflections	35
32	3.5 Discussion	36
33	3.6 Figure legends	39
34	3.7 Tables	40
35	4 CHAPTER 3	43
36	4.1 Introduction	43
37	4.1.1 Literature Review	44
38	4.2 Methods	46
39	4.2.1 Stimuli	46
40	4.2.2 Charts	49
41	4.2.3 Subject Recruitment	52
42	4.2.4 Experimental Design	53

43	4.3	Limitations	58
44	4.4	Conclusion	59

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48 This is the non-chapter stuff

49 2 CHAPTER 0

50 2.1 Motivation and Background

51 Data visualizations play an essential role in understanding patterns in the structure of
52 data. They allow for a programmatic mapping of data into a “picture” that can be used
53 to gather insight from the viewer [@tukey1965; @tufte2001; @wilkinson2005]. Since the
54 early 20th century, researchers have been exploring the question: what defines a good
55 graph? [@croxton1927; @croxton1932; @cleveland1984; @vanderplas2020]. Many attempts
56 to answer this questions have provided good recommendations, but are largely limited to
57 the projection of graphs onto 2-dimensional (2D) surfaces. While this addresses many of
58 the typical use cases for data visualizations, graphs created in our 3-dimensional (3D) world
59 are largely assessed.

60 Many charts exist within the confines of a 2D space due to the practicality of computer ren-
61 derings. These computer-generated graphics are quick and cheap to produce [@tukey1965],
62 which possibly contributes to their widespread usage. Before computers, hand-drawn and
63 complex printing techniques were common production methods and took time to produce
64 [@friendly2021; @vanderplas2019]. In contrast, all data physicalizations require additional

65 material resources, increasing the base cost to produce. This results in the majority of
66 modern graphics being produced with 2D representations on digital screens.

67 Nearly every type of chart can be created by using a programmatic mapping of data and vari-
68 ables to aesthetics and geometries, a process @wilkinson2005 referred to as the “Grammar
69 of Graphics.” This has found widespread implementation into numerous software [@wick-
70 ham2010; @satyanarayan2017; @sas9.4]. For example, consider the chart created in Figure
71 Figure 2.1. In this figure, the Palmer Penguins dataset [@penguins] is supplied to the
72 `ggplot` function from the `ggplot2` package [@ggplot2]. The x and y axes are mapped to
73 bill length and bill depth, with color and shape being mapped to island. The geometry is
74 given by `geom_point`, which results in a scatter plot that shows each complete data observa-
75 tion. Although the chart would benefit from further customization, it was quick to produce
76 and has instant visual insights.

```
library(tidyverse)

library(palmerpenguins)

ggplot(penguins, mapping = aes(x = bill_length_mm,
                               y = bill_depth_mm,
                               color = island,
                               shape = island)) +

  geom_point()
```

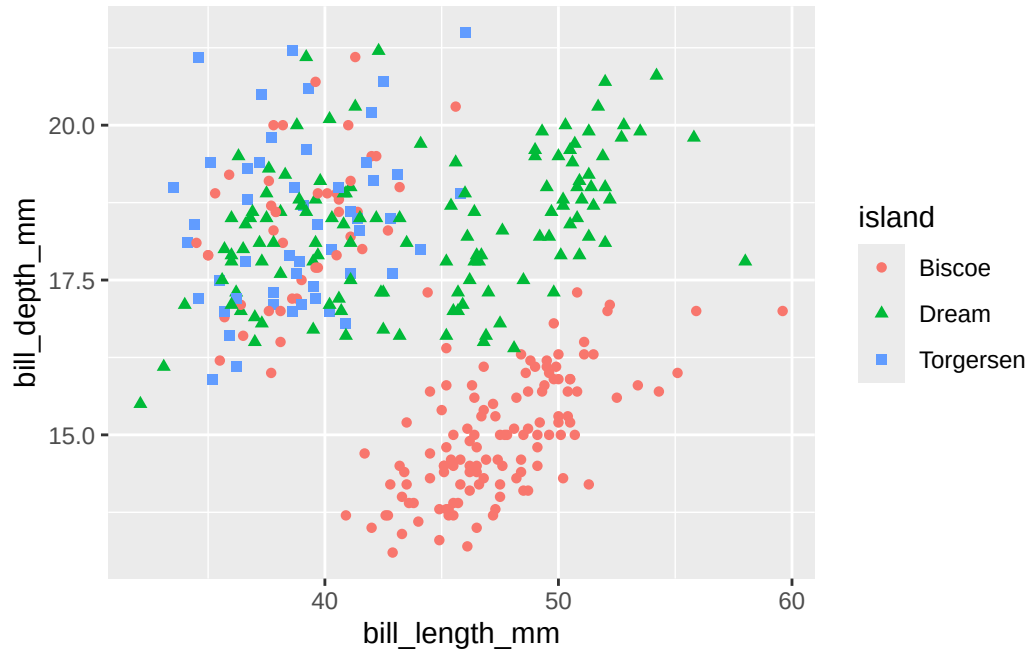


Figure 2.1: Example of a graphic created using the grammar of graphics.

While there have been many attempts to establish good data visualization practices [tuft2001; vanderplas2020], we do not know how well these recommendations translate from 2D projections into our 3D world. The advent of 3D printing allows us to precisely construct the 3D charts that have widely been studied through their 2D representations [croxton1932; barfield1989; zacks1998]. With limited research on 3D-printed data manifestations, we will explore the initial empirical findings of true 3D statistical graphics.

2.2 Overview of Physical 3D Visualizations

2.2.1 Creating 3D-Printed Visualizations

The popularity of the 3D printer exploded in the early 2010's as the technology became increasingly available and cheaper. This has led to increase usage of 3D printing in many

88 focus areas, including healthcare [dodziuk2016] and engineering [v2023], but has only
89 seen novel use cases for statistical graphics. Many software programs have the ability to
90 create 3D statistical graphics, but lack the ability to easily export the graphics into files
91 suitable for 3D printing.

92 Numerous software programs have the can create renderings of 3D charts. Excel [msexcel]
93 has native support for creating charts with 3D depth cues, but requires add-ins to produce
94 charts using 3 axes. R (XXX) has a several options for creating 3D charts that have data on
95 3 axes [e.g., rgl; rayshader; plotly], but lacks support for creating depth charts. SAS
96 9.4 (XXX) has options such as PROC GCHART for adding depth cues and PROC G3D
97 for surface charts. Other popular software programs contain similar capabilities, such as
98 JavaScript libraries, MATLAB, Python, and more.

99 While a number of tools exist for creating 3D data visualizations, the pipeline of getting
100 these charts into files compatible with 3D printing is not widely automated. For example,
101 R has some options using the `rgl` [rgl] and `rayshader` [rayshader] packages, but both
102 packages have limited tuning parameters for the resulting output files. Another option is to
103 use 3D software, such as OpenSCAD [kintelOpenSCADDocumentation2023], to manually
104 create the charts, but these software programs lack the ability to directly integrate statistical
105 information.

106 Another consideration for 3D-printed charts is the inclusion of text labels. With single-
107 filament 3D printers, text can be incorporated in three primary ways: (1) embossing, in
108 which the text is raised above the surface of the print; (2) engraving, in which the text is
109 recessed into the surface; or (3) applying external labels, such as stickers or ink. Multi-
110 filament 3D printers have the additional option to use a different color to make text labels

111 flush with their surfaces. However, @munzner cautioned that the text of 3D charts will
112 suffer from legibility issues due to the distortion of the text when viewed at an angle.

113 FIGURE PLACEHOLDER: examples of implementations of text on the charts

114 2.2.2 Current use cases

115 The use of 3D-printed data visualizations for statistical graphics is largely a novelty in the
116 early 21st century. Services such as WhiteClouds (www.whiteclouds.com) offer the ability
117 to create a wide variety of 3D-printed graphics, and more “do-it-yourself” options can be
118 achieved with \$300-\$500 in start-up costs. After acquiring a 3D printer, 1 kilogram rolls
119 of PLA filament cost approximately \$10 on Amazon, resulting in relatively inexpensive
120 charts.

121 FIGURE PLACEHOLDER: show some examples of 3D printed charts

122 2.3 Bar Charts in Research and Practice

123 One of the most common types of charts is the bar chart, characterized by the use of
124 rectangular geometries. Two variables are mapped to the geometries of the bar chart,
125 where one axis is dedicated to a categorical variable of nominal or ordinal nature, and the
126 other axis is for a continuous variable to denote magnitude or a response. These charts
127 have been used over 200 years, credited to William Playfair in his visualization of trade
128 data [@friendly2021].

129 The use of 3-dimensional elements in bar charts typically involve converting the rectangu-
130 lar bars into rectangular prisms. The naming convention for these types of charts is not

131 consistent, sometimes referred to as 2.5D charts when the additional dimension does not
132 convey extra information [e.g., @tractinsky1999] or, more commonly, as 3D bar charts [e.g.,
133 @zacks1998; @fischer2000]. All following references to this style of chart will be called “3D
134 bar charts.”

135 **2.3.1 Against 3D Bar Charts**

136 @tuftes2001 called the use of 3D elements a “fake perspective” in his description of “chart
137 junk”, which is where visual elements add clutter the data visualization. Other researchers
138 [@zacks1998; @stewart2009] have termed the perspective effect “extraneous” when referring
139 to depth cues. The reasoning behind this is fairly intuitive: why include additional noise
140 when simpler alternatives exist?

141 The seminal work of @cleveland1984 theorized that encoding information in volumes would
142 provide worse numerical estimation than for encodings using positions, lengths, and areas.
143 Their reasoning stems from Steven’s Power Law (XXX), a psychophysics formulation that
144 estimates a perceived stimuli magnitude p with the actual magnitude a by $p = ka^\alpha$. The
145 estimation of α provides guidance to what types of stimuli are subject to the most distorted
146 to their true scale when $\alpha = 1$. (alpha for volume estimated to be X, area is Y, and length
147 is Z). While the 3D bar charts do not lose their 2D encodings, they do gain depth, and
148 thus volume encodings. The effect of volume versus area comparisons was noted earlier by
149 @croxton1932, where metrics of accuracy were the worst for cubes as compared to bars and
150 circles.

2.3.2 For 3D Bar Charts

@levy1996 made an argument for the use of depth cues in certain situations. In a series of three experiments, @levy1996 tasked over 100 psychology students with selecting one of various 2D and 3D charts that they thought would be best for certain scenarios. What they found was that students tended to pick 3D charts over 2D charts when needing to increase memory recollection of the chart, and about equal preference for 2D and 3D charts when presenting the information to others. @levy1996 theorized that the depth embellishments in the graphic may increase memorability of the chart by making the chart stand out.

Other types of “chart junk” have shown that additional non-data features can increase recollection of bar charts at the cost of needing longer processing times [@bateman2010; @borgo2012; @peña2020]. This increase in time has been shown for 3D bar charts [@siegrist1996; @fischer2000; @stewart2009], lending some credibility to the argument that 3D depth cues could increase memorability.

2.3.3 Conflicting Empirical Studies

When accuracy is the goal, 2D is not better than 3D in some cases

Although @tufte2001 argued against the use of 3D charts when possible, empirical studies involving direct comparisons of 2D and 3D bar charts have shown mixed results. In general, these studies focus on metrics of accuracy and find that 3D bar charts tend to be either less accurate or just as accurate as their 2D counterparts. Another common metric is response time, where 3D charts tend to have longer response times. Overall preference for one style of chart over another is a subjective measurement, but has shown mixed results for the two styles of charts.

2.3.3.1 Accuracy

Accuracy is a common metric in studies comparing 2D and 3D bar charts, typically measured by either extracting a single numeric estimate or making comparisons between two values. Many studies used controlled environments for constructing stimuli, often displaying one or two stimuli at a time, or controlling non-target stimuli. In practice, charts are often more complex, displaying many stimuli without drawing attention to particular values.

When only measuring accuracy, 3D bar charts tend to be less accurate than their 2D counterparts [zacks1998; stewart2009]. However, this result is less prominent when introducing time delays [zacks1998, Experiment 2] or easier task complexity [stewart2009]. @siegrist1996 noted that bar positions and sizes also affected performance accuracy, emphasizing that other factors contribute to accuracy. It is also important to note that metrics of accuracy are not conclusive, where sometimes there is no evidence of a difference between 2D and 3D bar charts [melodycarswell1991].

Of course, there are other less-studied factors that could affect perceptual judgments of accuracy in 3D bar charts. In addition to viewing parameters found in 2D bar chart studies [fischer2005; rice2024], adding depth cues are also subject to viewing angles, amount of depth, and field of view parameters. Viewing angle, with respect to the axis, can mislead the reader into overestimating or underestimating numerical quantities (FIGURE X). @zacks1998 did not find a significant effect for the amount of perception depth, but noted a trend such that increased exaggeration of depth lowered their accuracy metric. Lastly, field of view is fixed for the viewer, but can widely distort 3D renderings (FIGURE X). These factors add additional complexity, but are not widely studied for the use of statistical graphics.

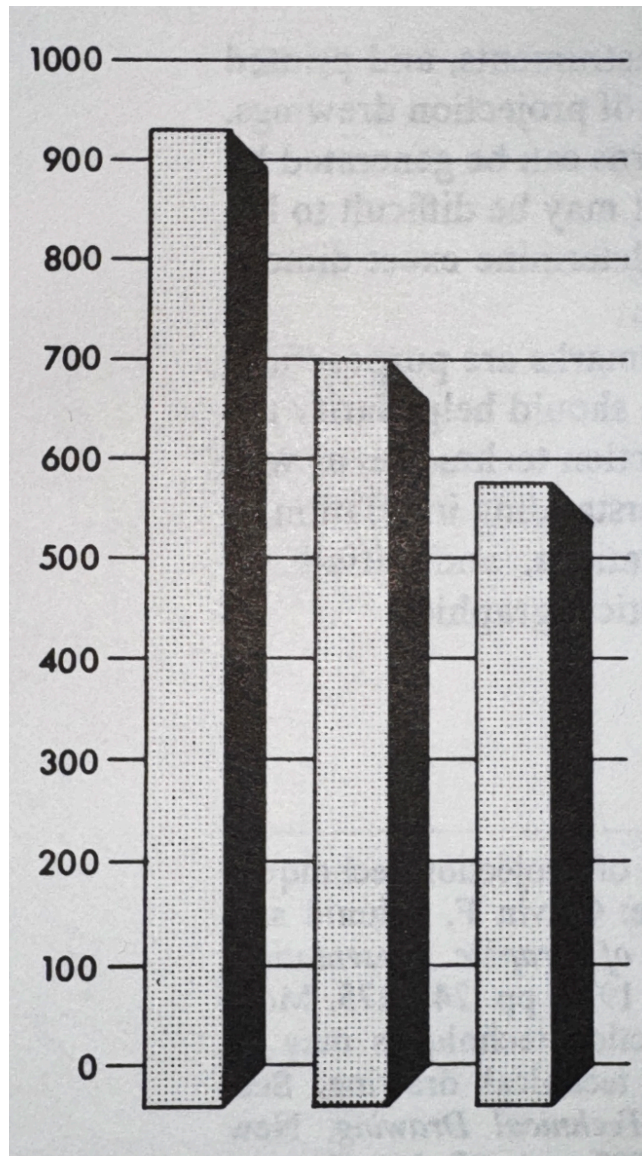
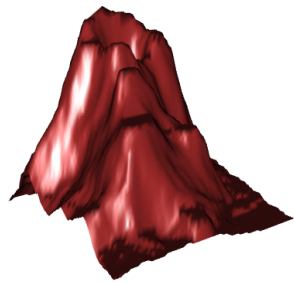


Figure 2.2: Figure in Statistical Graphics: Design Principles and Practices. The viewing angle for this chart would have viewers overestimate the values of the bars

196 2.3.3.2 Response Times

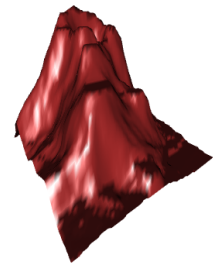
197 The speed in making perceptual judgments is another common metric in studies comparing
 198 types of charts. In theory, charts where questions can be answered more quickly implies
 199 that the chart is better at communicating information. However, response time is a poor



(a) FOV: 30 degrees



(b) FOV: 60 degrees



(c) FOV: 90 degrees



(d) FOV: 120 degrees



(e) FOV: 150 degrees

Figure 2.3: Multiple charts of the volcano dataset [`@volcano`] rendered using the `rgl` package [`@rgl`] in R. Each chart has a different field-of-view parameter, with the amount of distortion increasing as the field-of-view (FOV) increases.

metric in situations of exploratory data analysis and long term interactions with complex graphics [vanderplas2020]. When a question of interest is known in advance, response time becomes a more viable metric.

Depth cues from 3D bar charts tend to increase the amount of time required to answer questions [siegrist1996; fischer2000; stewart2009]. Intuitively, additional complexity should require additional time to process. For 3D bar charts, viewers need to align the response axis with the rectangular prisms in order to determine the approximate magnitude of the bar. Consider Figure X, where the y-axis denotes magnitude with respect to a hidden face of the bar. In order to extract a numerical quantity, the viewer first has to align the axis with the correct face of the prism before processing the magnitude.

2.3.3.3 Preference

The debate between whether to use 2D or 3D graphics stems from a wider adoption of 3D charts in practice. Computer graphics have made it easy to produce charts of either dimensionality, resulting in creative liberties for the creation of the chart for publication. Figures X and X showcase two examples of 3D bar charts from the 1978 Handbook of Agricultural Charts (XXX) where other options exist for displaying the data. Schmid1983 (Ch. 8) also showcases many other 3D charts published in official reports.

U.S. SUGARBEET AND SUGARCANE PRODUCTION

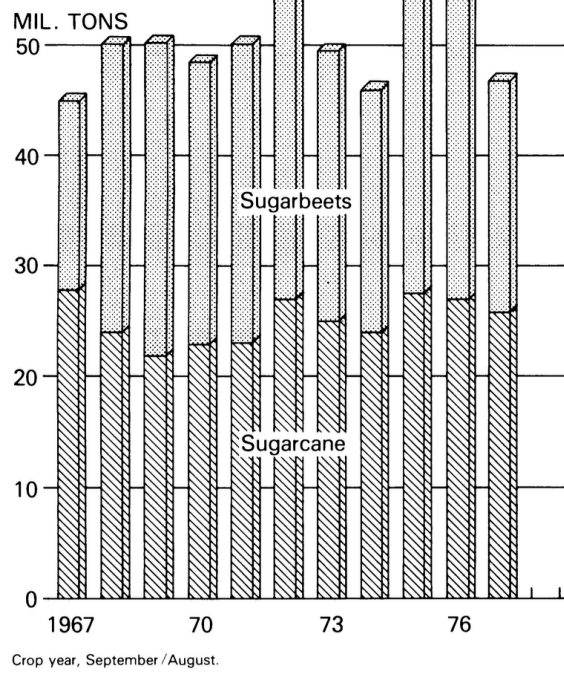
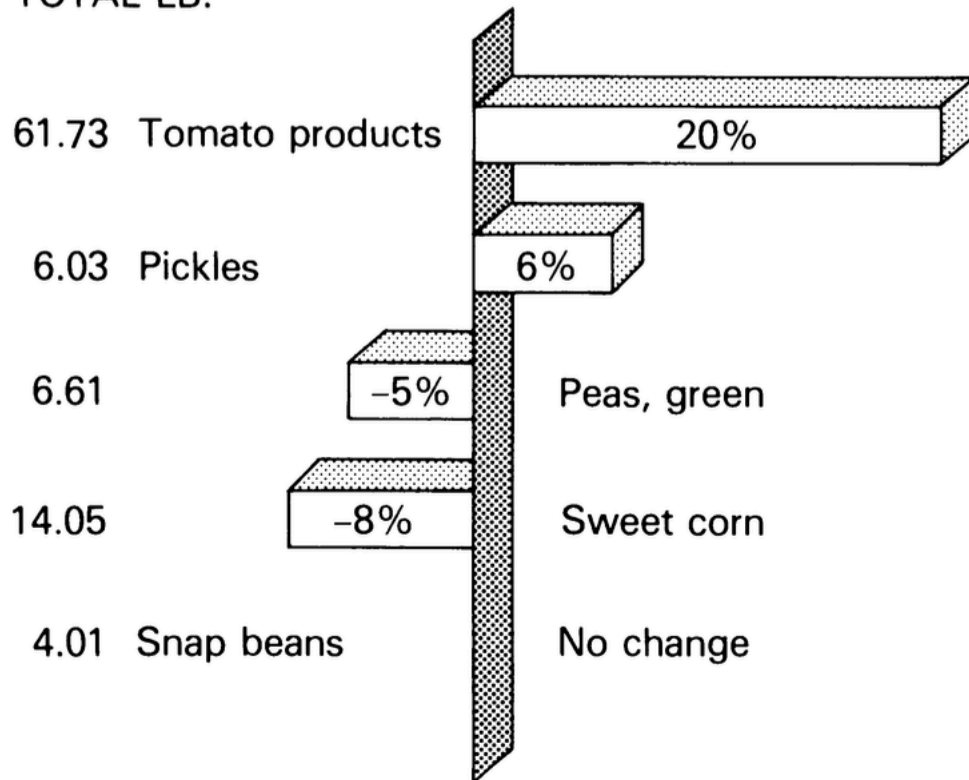


Figure 2.4: Located from 1978 HANDBOOK OF AGRICULTURAL CHARTS

CHANGES IN CANNED VEGETABLE CONSUMPTION PER CAPITA BETWEEN 1970-72 AND 1975-77

TOTAL LB.



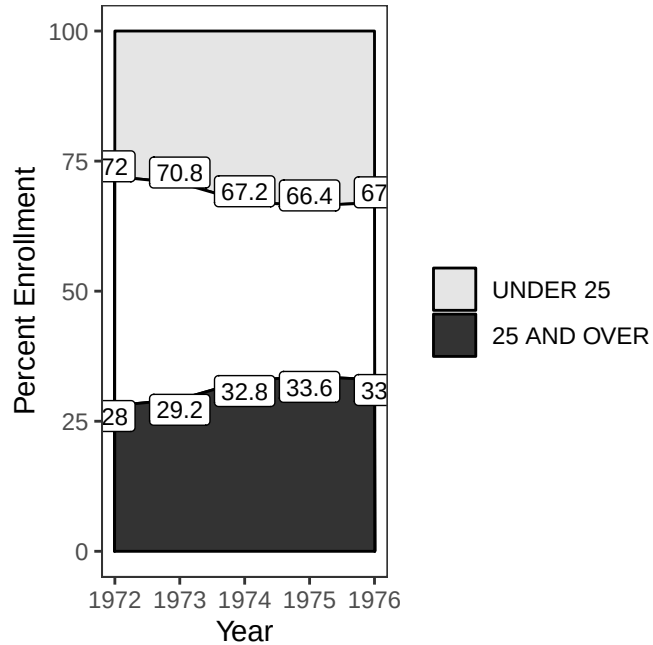
Fresh-weight basis.

217

218 Preference can be broken down into two broad categories: (1) visual appeal, and (2) useful-
 219 ness. Visual appeal is where perceptual tasks are not a priority, and the goal is to have an
 220 image that is pleasant to the viewer. Usefulness, on the other hand, is a rating of how effec-
 221 tive the viewer thought the chart was in order to answer the question. Both are subjective
 222 measures

223 When not tasked with estimating values, 3D bar charts tend to be more visually appealing
 224 [Levy1996; Fisher1997; Stewart2009]. This may be in part due to the forced illusion
 225 of depth which creates additional embellishments that would increase the impression and

memorability of the chart, as seen by other types of Tufte’s “chart junk” [borgo2012;
 @peña2020]. Despite the visual appeal, 2D charts tend to be chosen for preference and ease
 of use when making numerical estimations [levy1996; @stewart2009].



2.3.4 Construction of 3D Bar Charts

All of the aforementioned studies have one key limitation: they are conducted in environ-
 ments where the 3D bar charts are projected onto 2D surfaces. An argument could be
 made that tactile charts created for persons with vision impairments could be considered
 3D [goncu2008; engel2017; @watanabe2018], but current implementations are vastly
 different from the typical design of 3D bar charts.

2.4 Heat Maps in Research and Practice

When creating charts for three or more variables, it becomes increasingly complex to vi-
 sualize data [grinstein2001]. One option is to map the additional variables using gestalt

principles [vanderplas2020] (Original citation?), which includes making use of color, size, and/or shapes to denote information. However, not every aesthetic is compatible with every type of variable. Color can be used to map either continuous variables with gradient scales or categorical variables with palettes. Size is typically reserved for continuous variables, where larger values correspond to a larger quantity. Shapes tend to have finite distinguishable representations in order to represent categorical variables. Examples of good and poor cases for each of these aesthetics is presented in Figure Figure 2.5.

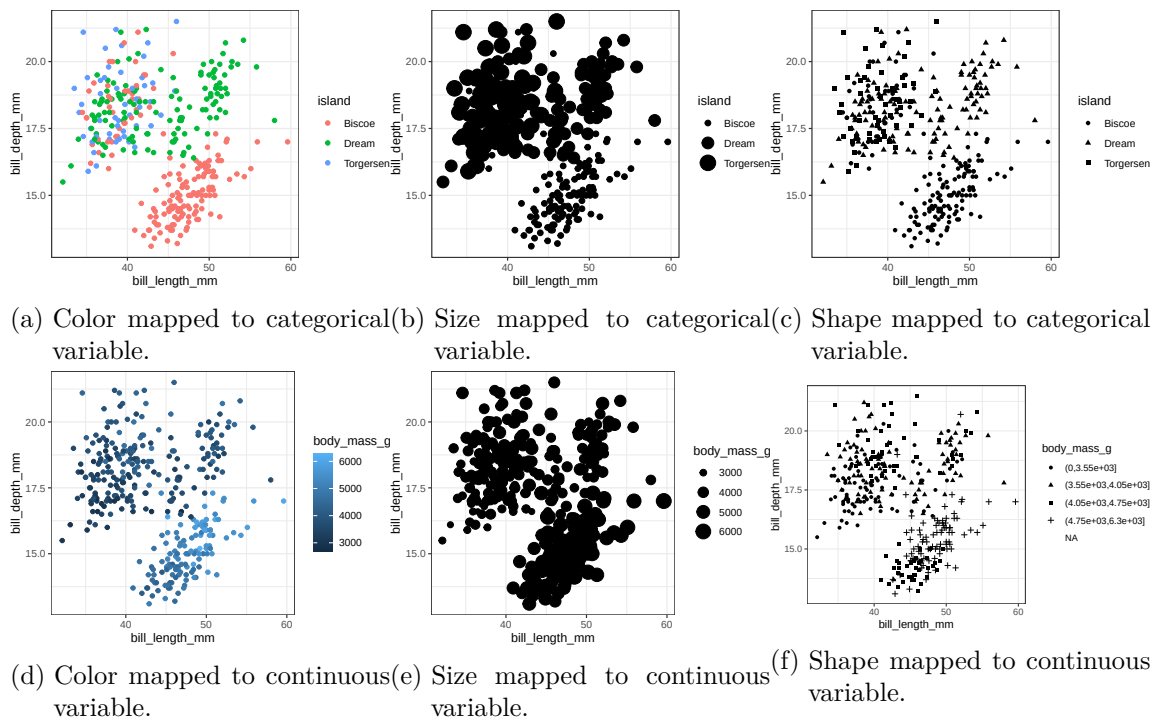


Figure 2.5: Examples of color, size, and shape for scatterplots of the penguins dataset [pen-guins].

Heat maps are a type of chart that can be used with two variables of any type and a continuous response variable. In two-dimensional formats, color is often used to denote the magnitude of the response, utilizing a gradient scale. When represented in three dimensions, height is the primary mapping of the response of the heat map. Due to the height aesthetic, 3D heat maps are sometimes referred to as “height maps”, particularly when

251 spatial coordinates are involved.

252 **2.4.1 Empirical Studies of 2D and 3D Heat Maps**

253 Unlike 2D and 3D bar charts, the comparison of dimensionality for heat maps has not been
254 widely studied. This may in part be due to the complex nature of color scales [@breslow2009;
255 @liu2018; @reda2019]. While bar charts simply add a perspective depth cue to increase the
256 dimensionality, the translation of color to height results in numerous considerations, such
257 as the scaling of the height (Figure @fig-color-height).

258 The type of media can affect the perception of 2D and 3D heat maps [@kraus2020;
259 @jeong2025].

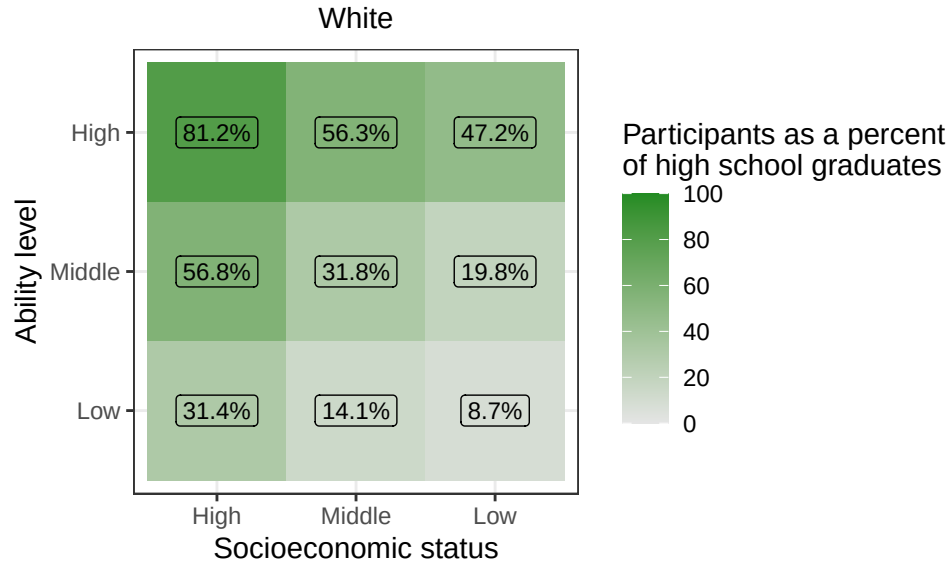
- 260 • @barfield1989
- 261 • @kraus2020
- 262 • @jeong2025 would be good, but uses VR
- 263 • @casali1988

264 **2.5 Methods for Evaluating Visualization Effectiveness**

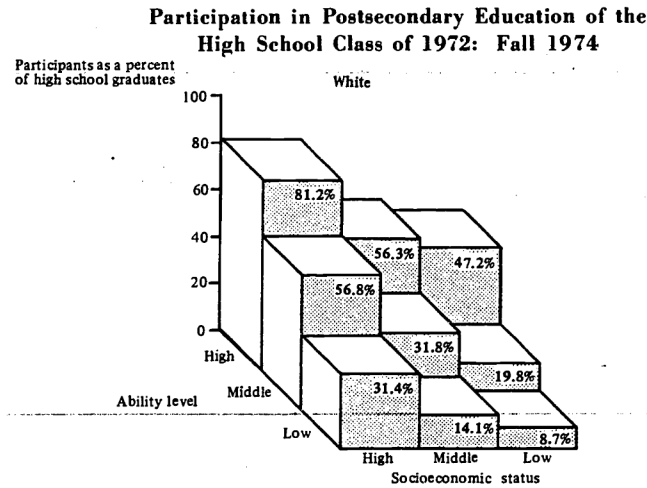
265 **2.5.1 What makes a good chart?**

266 @vanderplas2020 discussed various approaches to testing graphics, including numerical esti-
267 mations, speed, and error rates. These metrics are fairly common metrics when comparing
268 2D and 3D charts [@croxton1932; @cleveland1984; @hughes], but do not represent the full

Participation in Postsecondary Education of the High School Class of 1972: Fall 1974



(a) 2D heat map



(b) 3D heat map

Figure 2.6: Heat maps created from with data from (XXX). The 2D heat map represents the percentage of high school graduates using a gradient color scale, whereas the 3D heat map uses height.

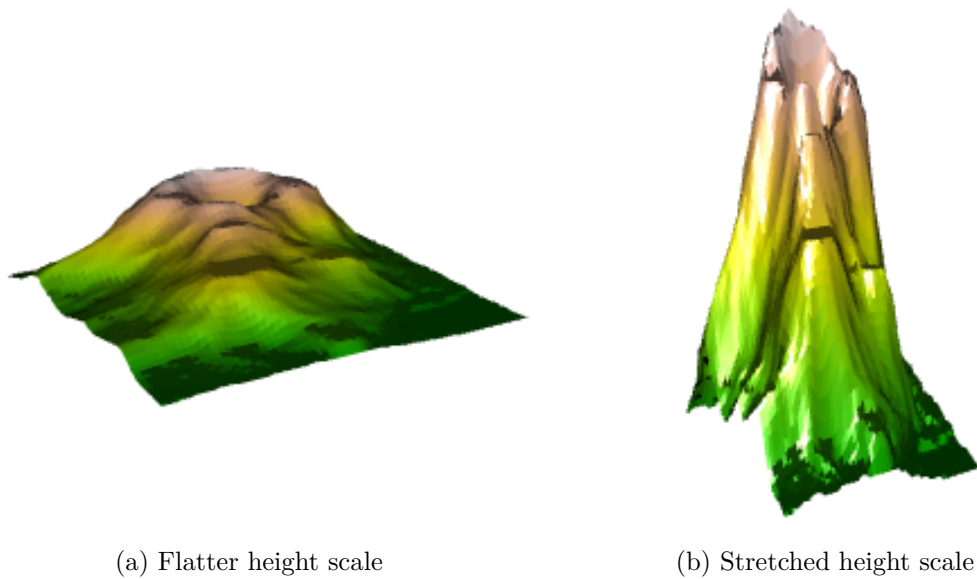


Figure 2.7: Two figures representing the volcano dataset [volcano] using different height scaling factors. Both figures use the same color scale, but there is no “correct” translation of color into the measurable height.

picture of what makes a one type of chart better than another. As seen by @levy1996, the objectives of the chart can determine which type of chart should be used.

Since true 3D charts have limited research compared to 2D representations, it is reasonable to compare how perceptual judgements of numerical estimations compare to earlier studies in statistical graphics.

2.5.2 Designing studies for numerical estimations

Over 2,500 citations have referenced the work of @cleveland1984 when discussing properties of statistical graphics. Since then, the methods of analyzing and conducting studies on statistical graphics have evolved [e.g., @vanderplas2020; @hofmann2012], but yet the structure of experimental designs involving numerical estimations tend to remain similar. Tasks typically involve the comparison of two stimuli, where the question is “how much

280 bigger/smaller is one value compared to the other?” Care needs to be taken in the phrasing
281 of this question since mental subtractive processes are perceived different from mental ratio
282 processes [@veit; @hagerty1978].

283 @cleveland1984 asked participants two questions when making comparisons between two
284 values on a chart: (1) which of the two values was the smaller, and (2) what percentage the
285 smaller was of the larger. The first question clarified if the participants were looking at the
286 values correctly for the second questions, and the second question is an estimation of A/B ,
287 where $A < B$. Accuracy was measured by $\log_2(|\text{True value} - \text{Estimated value}| + 1/8)$, using
288 the trimmed averages when bootstrapping confidence intervals. Cleveland and McGill’s
289 experiment was replicated by @heer2010, showing similar results for the experiment’s pro-
290 cedure in a crowdsourced environment.

291 When designing studies for numerical estimation, the choice of A/B and the measure of error
292 are important considerations. Many researchers seemingly choose ratios in an arbitrary
293 fashion or follow previously established work [e.g., @zacks1998 used the same ratios as
294 @melodycarswell1991]. As for measures of error, several formulations can be used.

- 295 • $|\text{Error}|$ for magnitude of error
- 296 • Error for direction of error
- 297 • $\log |\text{Error}|$

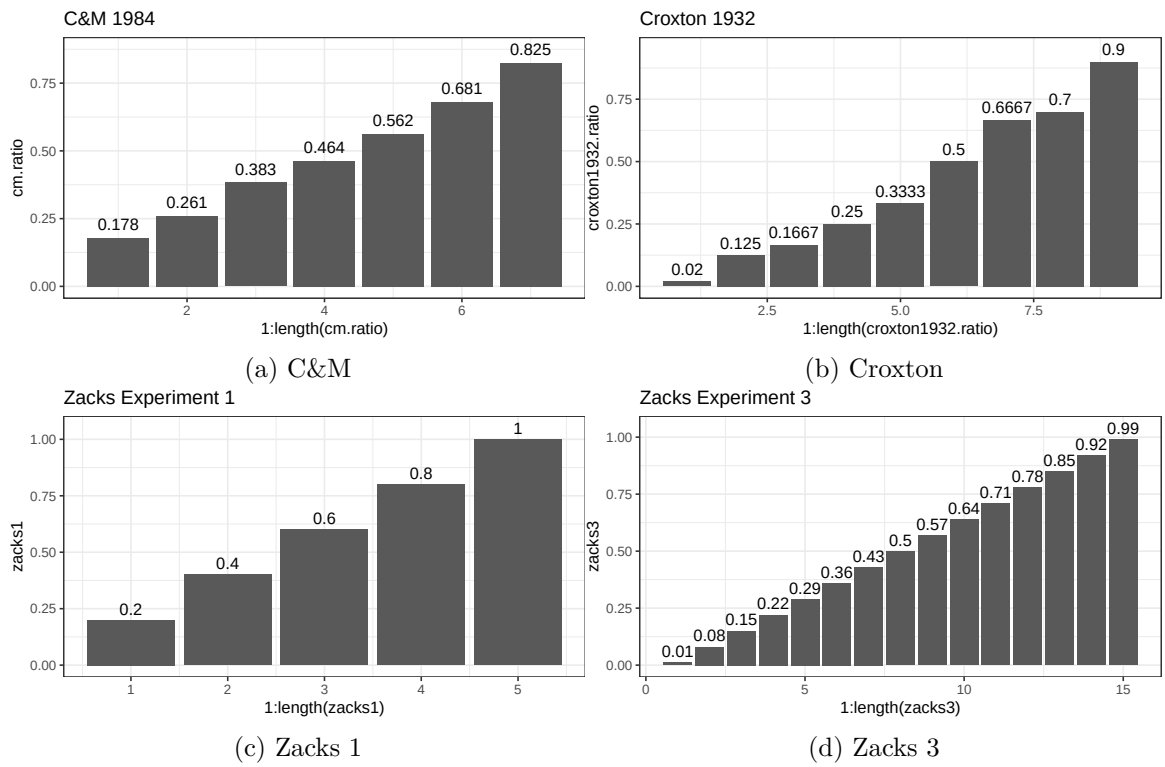


Figure 2.8: Ratios used by several studies

298 3 CHAPTER 2

299 3.1 Abstract

300 A key component of statistics courses is to teach students how to interpret data visual-
301 izations. Although many research-based recommendations exist for creating graphs, the
302 technological advances for creating such graphs have outpaced studies that evaluate their
303 effectiveness, especially when considering 3D graphs. Here, we describe a process for inte-
304 grating an experiment on 3D graphs as part of a project on statistical investigations for
305 students enrolled in an introductory statistics course and gathered responses as students
306 reflected on their experience of being a participant in the experiment and then being a
307 reviewer of empirical evidence about the experiment A total of 82 students participated in
308 our graphics project. As participants in an experiment, they displayed a pattern of not fully
309 grasping research objectives as evidenced by widely varied responses. However, as reviewers
310 of material from a pilot study of the same design, they tended to gain a clearer understand-
311 ing of the purpose of the experiment and its role in the realm of data visualizations by
312 correctly interpreting an extended abstract and video presentation of the pilot study. The
313 project we presented to students shows promise as an educational tool for helping students
314 gain a more holistic view of statistical research, which is important in both the contributions

315 to data visualizations and the education of statistics.

316 **3.2 Introduction**

317 The education of statistics emphasizes the use of real data and its application in answering
318 research questions. Students in introductory statistics courses are mainly exposed to elemen-
319 tary methods and textbook examples that demonstrate the application of these methods,
320 which is used, in part, because of the emphasis on teaching students to think statistically
321 using real data [carver]. Many textbooks, such as Tintle2021, use scenarios and ask
322 students to perform the corresponding inferential test. This process is then repeated over
323 the course material without much deviation in the instructional method. In some cases,
324 however, students can participate and benefit from well-designed classroom experiments
325 [mcgowan2011]. This process can expose students to concepts such as randomization and
326 let students see the specifics of experimental design through their participation. loy2021a
327 has demonstrated that student participants often recalled their experiment in later concepts,
328 showing some evidence that students can benefit from the hands-on experience.

329 In addition to using experiential learning broadly, a key aspect in the statistics classroom
330 is teaching students to interpret data through graphs. Nearly 40 years ago, cleveland1984
331 began the process of establishing good practices for making graphs. While their findings
332 have been replicated [heer2010], many areas in data visualization remain underdeveloped
333 that can benefit from the framework used by cleveland1984. One such area is the projec-
334 tions of 3D data visualizations. The current mantra is to avoid 3D graphs when possible and
335 studies around the 1990s seem to provide some valid skepticism of their use [barfield1989;
336 fisher1997]. barfield1989 showed that participants were sometimes more accurate us-

ing 3D graphs than 2D graphs, depending on the participant’s experience level. However, participants were generally more confident with their answers for 2D graphs. @fisher1997 observed a similar preference for 2D graphs over 3D graphs when extracting information, but found no preference for visual appeal for the type of graph. While these results provided valid skepticism toward the use of 3D graphs, the results are generalized to the projections of the 3D graphs. The area of “true” 3D graphs is largely unexplored, but advances in technology allow for easier access to explore the 3D projections of these graphs. @kraus2020 explored the 3D projections with the use of virtual reality and found that participants were more accurate at extracting values from 3D heatmaps at the expense of needing more time than 2D alternatives.

Because the use of “true” 3D graphs remains largely unexplored, it provides a unique opportunity to be used as an experiential learning opportunity for statistics students. Not only can students benefit from the exposure to different graph types, including “true” 3D graphs, but they can also experience a more authentic method of teaching, which is more likely to be beneficial to reinforce statistical ways of thinking.

In this paper, we used an experiential learning project that employed different graph types, including “true” 3D graphs, in an introductory statistics classroom environment and described its potential application as an educational tool. We presented students with a series of modules where they first participated in our experiment on different graph projections, followed by acquiring knowledge on the purpose of the experiment by reading an extended abstract and watching a presentation of a pilot study of the same experiment.

3.3 Methods

3.3.1 Overview and Motivation

We introduced students enrolled in Introduction to Statistics (STAT 218) at the University of Nebraska-Lincoln to a graphics project that contained an experiment and progressively revealed components that illustrate the experiment’s research objectives. The project started by providing students with minimal information about the research objectives before revealing the scope of the experiment through an extended abstract and presentation. The goal of the graphics project was to observe how students think statistically about experiments as both participants and research consumers. Students were provided with mostly open-ended prompts throughout the graphics project, which allowed us to freely explore common themes in student responses without the limitation of preset choices.

3.3.2 Experiential Learning

The graphics project was split into two main components regarding the interaction of students with the material. In this section, we will discuss student interactions with the experiment. Students were first presented with a series of four modules presented from the role of research participation. These modules contained the informed consent form, pre-experiment reflection, experiment participation, and post-experiment reflection. Within the informed consent module, students were informed that their data would be anonymized and that the experiment was exempt from the institutional review board (Project ID: 22579). While all students were given the option to participate in the graphics project, we were only able to collect responses when informed consent was obtained and if the student was

19 years of age or older. In the pre-experiment reflection, students were asked to write a paragraph about how they think the process of scientific investigation looks from the perspective of researchers and the general public. The experiment participation module asked students to paste the code generated upon completion of the graphics experiment, which is detailed in the next section. The generated code serves as a basic check to indicate whether or not students fully participated in the experiment. For the post-experiment reflection, students were asked five questions about the purpose of the experiment. These include questions on the hypotheses being tested, sources of error, variables of interest, and elements of experimental design.

After completing the experiment reflections, students moved to the reflection of the overall research objectives. Students were first directed to read an unpublished two-page extended abstract that we submitted as a contributed paper for the Symposium on Data Science & Statistics (SDSS). The extended abstract outlined the experiment’s purpose and procedures, but not the results from our initial pilot study. After reading the extended abstract, students were asked to write a paragraph about what they found clearer about the experiment’s purpose than when they were a participant. Finally, students were directed to watch a 12-minute pre-recorded presentation based on an abbreviated version given at SDSS [wiederich2023]. The video contained the same material as the extended abstract and included the results from our pilot study. The presentation reflection asked students four questions about the experiment and how information was presented differently than in the extended abstract.

Except for the informed consent and experiment participation modules, all student responses were open-ended. Each question and its corresponding module can be found in Table 1.

Instructors for STAT 218 were recruited for Summer 2023 and Fall 2023 to administer the graphics project into their classroom. The instructors were given the option of administering the project as coursework material or extra credit, along with the liberty of grading at their discretion. Instructors for the June-July Summer 2023 courses did not have the abstract or presentation modules.

3.3.3 Data Analysis

Since nearly all of the student responses to the project modules are open-ended, the analysis of the project modules is qualitative. We will selectively extract student responses that demonstrate variability and repetitive themes in their understanding of the graphics experiment. For all modules, except the Post-Experiment reflection, students are asked to comment on the experiment without an objectively correct response.

For paragraph responses in the Pre-Experiment and Abstract reflections, bigram plots are used as a graphical analysis to display word pairs after removing stop words (e.g., “the” and “and”). These word pairs help illustrate common themes that students wrote about in their longer prompts. In the Post-Experiment reflection, the prompts have objectively correct answers but would require a subjective aggregation to indicate the level of correctness. Instead, we opt to select at least one student response that we consider to be “most correct” and a couple of other responses that are either partially correct or entirely incorrect to illustrate the variability of the responses. The results of the graphics experiment are presented by @wiederich as an extension of a larger study comparing the dimensionality and projections of 2D and 3D bar charts.

423 3.3.4 Graphics Experiment

424 3.3.4.1 Constructing Stimuli

425 Based on Cleveland and McGill’s seminal work [–@cleveland1984] on graphical perception,
426 participants were presented with a series of bar graphs where two bars are marked with either
427 a circle or a triangle. The heights of the bars were chosen from the following equation:

$$s_i = 10 \cdot 10^{(i-1)/12}, \quad i = 1, 2, 3, \dots, 10 \quad (3.1)$$

428 where s_i represents a value given an integer i as defined above. The values s_i from Equa-
429 tion 3.1 were then paired such that the ratio of the smaller value to the larger value yield
430 the ratios of 0.178, 0.261, 0.383, 0.464, 0.562, 0.681, and 0.825. Each bar graph has two
431 groupings of five bars. Following the Type 1 and Type 3 graphs from the position-length
432 experiment by @cleveland1984, the value pairs for each ratio were either placed in the first
433 grouping on the second and third bars (adjacent) or placed on the second bars in each
434 grouping (separated).

435 To explore the effect of dimensionality and projection of the bar graphs, we introduced the
436 following plot types: 2D digital, 3D digital (static), 3D digital (interactive), and 3D printed.
437 There was no single software package that could create all four plot types, so we carefully
438 constructed graphs from different software packages to be as similar as possible. The 2D
439 digital plots were rendered with the `ggplot2` package [–@ggplot2]. Microsoft Excel® was
440 used to render the 3D digital (static) plots. The 3D digital (interactive) and 3D printed
441 plots were created with OpenSCAD® [–@kintelOpenSCADDocumentation2023], where the

442 generated stereolithography (STL) files for the 3D digital (interactive) plots were rendered
443 with the `rgl` package [Rgl].

444 **3.3.4.2 Experimental Design**

445 With 56 treatment combinations, we opted to use an incomplete block design to provide
446 participants with 15-20 graphs. Kits of graphs were constructed so that five of the seven
447 ratios are equally represented, resulting in 21 different kits. Within each kit, all graph types
448 appeared for each ratio and the comparison type was randomly assigned. A visual layout
449 of the experiment is shown in Figure 1. All kits received a unique identifier and a set of
450 instructions for accessing the experiment.

451 A Shiny application [Shiny] was designed to administer the experiment. Students were
452 directed to randomly select a kit of graphs and visit the Shiny application's website linked
453 on the instructions. For students enrolled in the online sections of STAT 218, the website
454 was provided by the instructor and they were prompted in the application to select that
455 they were an online participant; selecting online participation resulted in the 3D-printed
456 plots being removed from the set of graphs presented to the participant. After students
457 provided a kit identifier (if applicable), students were presented with graphs in a randomized
458 order. If the student marked that they were an online participant, the 3D-printed graphs
459 were removed from their experiment lineup. Each graph asked the students to first identify
460 the larger marked bar and then to guess the height of the smaller marked bar if the larger
461 marked bar was 100 units tall using a slider widget. After completing the experiment,
462 a completion code was generated for students to paste into the experiment participation
463 module.

464 **3.4 Results**

465 **3.4.1 Recruitment of Students and Instructors**

466 We recruited 3 instructors for the summer and fall semesters in 2023. Each instructor offered
467 the project as extra credit in their course and student participation was entirely voluntary.
468 A total of 82 students participated in the project, and 9 students did not complete the entire
469 project (Table 2).

470 **3.4.2 Selected Responses from Experiment Participation**

471 **3.4.2.1 Pre-Experiment Reflection**

472 In the first stage of the project, before participating in the experiment, students had limited
473 information about the research objectives and we recorded how students thought about
474 scientific research. Students were initially prompted with one question, “What do you
475 think [the] process [of scientific investigation] looks like, from the perspective of a researcher,
476 compared to what it looks like from the perspective of someone in the general public who
477 is a consumer of scientific results?” The responses to this question allowed us to gather a
478 broad understanding of what students initially thought about the research process. Students
479 generally understood the purpose of scientific research by connecting the ideas of hypothesis
480 testing and publishing results. Students wrote about scientific research starting from the
481 place of a question, followed by conducting an experiment and relaying the results to the
482 public. For example, the most common phrase groupings include variations of “scientific
483 research”, “data collection”, and “peer review”. A bigram plot of student responses to the
484 Pre-Experiment reflection prompt is shown in Figure 2.

485 3.4.2.2 Post-Experiment Reflection

486 After participating in the experiment, students were provided prompts asking about the
487 goals of the research objectives, which were to evaluate the errors of ratio judgments between
488 2D and 3D graph types under different comparison conditions in a randomized block de-
489 sign. Some students correctly identified parts of the questions asked in the post-experiment
490 reflection, but most students typically missed some or most of what would be considered a
491 correct answer.

492 We first asked students “What do you think the purpose of the experiment was?” For this
493 question, a complete response would have included the comparisons of ratio judgments of
494 the projections for 2D and 3D graphs. A student example of the correct response was “I
495 think this experiment aimed to test if it was easier to compare two graphs in 2D or 3D.”
496 Another student missed the comparison of 2D to 3D graphs, but was mostly correct with
497 their response “I think the purpose of this experiment was for the researcher to gather data
498 on how people perceive, interpret, and understand 3D graphs.” One student incorrectly
499 thought that we were testing differences in demographics by responding “They could be
500 trying to determine how different genders, ages, etc. perceive the sizes of the bars in the
501 graph. Demographics could make a pretty significant difference.”

502 We then asked students “What hypotheses might the experimenters have been testing?” A
503 correct response would include testing the differences in errors of ratio judgments between
504 graph types. One example of a student providing a correct response was given by stating
505 “They might have been testing if a 2D model is easier to estimate its relative size to another
506 when compared to a 3D model of it.” Other students replied with statements that would
507 not be able to be measured from the experiment, with one student responding “How taking

508 Statistics 218 effects how you can compare two groups” and another student saying “That
509 2d is preferred over 3d. It cleans up the data presentation.”

510 An important topic covered in STAT 218 is randomization, which we asked students with
511 the prompt “What elements of experimental design, such as randomization or the use of
512 a control group, do you think were present in the experiment? Why?” In the experiment,
513 randomization was performed with the ordering and assignment of graphs into kits. One
514 student perfectly described randomization by responding with “Randomization: The survey
515 used an experimental design where in the survey there were different sets of 3D charts and
516 maybe by a randomization process each participant was shown a different set of charts to
517 see the differences in interpretations of the charts based on which set was assigned. Control
518 Group: Since this survey aimed to only understand how participants interpret 3D charts
519 without comparison to other chart types, then no control group was needed.” Another
520 student was partially correct with a response of “Randomization was used because the ever
521 person got a different graph.” Other students missed the use of randomization, with one
522 student responding “random students in the stats class” and another saying “Randomization
523 was not used because it was offered as an extra credit assignment in class.”

524 **3.4.3 Selected Responses from Research Reflections**

525 In this set of reflections, students first read the extended abstract, followed by watching
526 the 12-minute presentation video. The abstract unveiled the scope of the study to students,
527 many of whom did not realize the underlying complexities. Nearly all students responded
528 with statements about gaining clarity about the purpose of the experiment and its role in
529 testing the differences between 2D and 3D graphs. Students commented on how they now

530 understood the purpose of comparing 2D and 3D graphs, and also the potential benefits
531 that may stem from research, such as graphical accessibility to the visually impaired. A
532 bigram plot of the student responses to the abstract reflection prompt is shown in Figure
533 3.

534 Lastly, more than half of the students (78.5%) responded that they preferred the presenta-
535 tion over the extended abstract when asked “If you had to hear about this study using only
536 the extended abstract or only the presentation, which one would you prefer? Which one
537 would be better for determining whether the experiment was well designed?” One student
538 responded “I am a visual learner so I would have rather heard about in through the pre-
539 sentation. It also broke down the steps which is easier for me to understand. I think the
540 presentation as a whole would be better for determining how the experiment is designed.”
541 Another student said “Personally I like the abstract better. If I get confused on something
542 it is so much easier to go back and reread to understand what is going on. If I ask myself
543 questions about it, it is much easier to go back and find answers to the questions as well.”

544 **3.5 Discussion**

545 Taken together, our results support the idea that we achieved our goal of providing stu-
546 dents of an introductory statistics course with the opportunity to reflect on active research.
547 Students generally appreciated the progressively revealing nature of the graphics project,
548 which is evident from the abstract and presentation reflections. When provided with the
549 post-experiment reflections, students often either missed the research objectives of the ex-
550 periment or had partially correct responses. However, this was expected since the design
551 of the experiment exceeds the scope of experimental design taught in typical introductory

552 statistics courses.

553 The abstract reflection received many responses indicating that students had moments of
554 realization about the true nature of our research goals, which was further expanded with
555 the presentation reflection prompts. Across all reflections, students were thoughtful, and
556 sometimes amusing, with their responses and they were on the path of statistical thinking.

557 Having students participate in experiments is not uncommon, possibly attributed to the
558 readily available convenient samples within academia and potential applications to introduce
559 new course material [margaret1994]. However, these studies typically end for students
560 after completing the experiment [mcgowan2011; zacks1998; fisher1997]. In our study,
561 we integrated the graphics experiment as a course project with the addition of reviewing
562 research material.

563 What we found was that the graphics project found success within the recommendations of
564 the GAISE College Report by using real data to illustrate the contextual purpose of scientific
565 research [carver]. Students were able to demonstrate their ability to think statistically
566 through the series of reflections. The graphics project made use of real data, along with
567 data collection, within the scope of an approachable and field-related topic, which allowed
568 students to see how scientific research is conducted in the field of statistics.

569 A limitation of our study is the use of open-ended responses that do not objectively assess
570 student learning. While the student responses were useful in gathering insight, the responses
571 are widely varied and do not have direct comparisons of statistical thinking throughout
572 the modules. Another limiting factor is the tiered layering of convenience sampling, with
573 instructors being recruited before recruiting students, which impacts the generalization of
574 our findings. Nonetheless, the 82 students who participated provided meaningful answers

575 that displayed some level of statistical thinking throughout the graphics project.

576 Future studies could use a similar framework to conduct experiments on more typical 3-
577 dimensional data, such as heatmaps. The use of graphical experiments in the classroom
578 not only provides a readily available convenience sample but also adheres to the recom-
579 mendations of the GAISE College Report [carver]. With the framework we provided in
580 this paper, we aim to make adjustments to further improve the graphics experiment and
581 corresponding project as an experiential learning opportunity.

582 3.6 Figure legends

583 Figure 1: Visual display of the experimental design for students who participated in the
584 3D bar charts experiment. Kits of graphs were created by first choosing five ratios from
585 nine available options (1). Each ratio then uses all graph types, with the exception of the
586 3D-printed graphs for online students (2). Finally, all graphs were randomly assigned to
587 have the marked bars as adjacent or separated (3).

588 Figure 2: Bigram of student responses (n=82) to the pre-experiment prompt. Each line
589 represents pairs of words that appeared together where each pair occurred at least twice.
590 Students generally understood that science is about investigating research questions and
591 collecting data.

592 Figure 3: Bigram of student responses (n=63) to the abstract reflection prompt. Each line
593 represents pairs of words that appeared together where each pair occurred at least twice.
594 Students generally focused on the research objective of comparing 2D and 3D graphs.

Table 3.1: Questions provided to students in each project module.

Reflection	Question	Prompt
Pre-Experiment	Q3	In this class, you'll be learning about the process of scientific investigation. What do you think that process looks like, from the perspective of a researcher, compared to what it looks like from the perspective of someone in the general public who is a consumer of scientific results? Write a paragraph (at least 3-5 sentences) about how you think science happens.
Post-Experiment	Q5	What do you think the purpose of the experiment was?
	Q6	What hypotheses might the experimenter have been testing?
	Q7	What sources of error are involved in this experiment?
	Q8	What variables were examined? For each variable, identify whether it was quantitative or categorical.
	Q9	What elements of experimental design, such as randomization or the use of a control group, do you think were present in the experiment? Why?
Abstract	Q10	What components of the experiment are clearer now than they were as a participant? What questions do you still have for the experimenter? Write 3-5 sentences reflecting on the abstract.

Presentation	Q11	How did the information you gained from the components of this project (participation, post-study reflection, extended abstract, presentation) differ?
	Q12	What components were emphasized in the presentation that weren't emphasized in the abstract? Why do you think that is?
	Q13	What critiques do you have of this study and its design? What would have made the study better?
	Q14	If you had to hear about this study using only the extended abstract or only the presentation, which one would you prefer? Which one would be better for determining whether the experiment was well designed?

Table 3.2: Number of valid student participants by semester.

Semester	Number of Sections	Number of Students
Summer 2023 (May-June)	1	17
Summer 2023 (July-Aug)	1	23
Fall 2023 (May-June)	1	42

Students under 19 years of age or did not consent were excluded from data collection. To comply with IRB, no demographic information was collected to keep students anonymous with their reflections.

596 4 CHAPTER 3

597 4.1 Introduction

598 In the 20th century, advancements in technology made data visualizations increasingly more
599 affordable and accessible to a broader population. The primary change in formal data
600 visualizations was from hand-drawn charts to computer-rendered charts [tukey1965], yet
601 other technological advances have allowed for data visualizations to enter other mediums.
602 These include the ability to effectively use the 3-dimensional (3D) world around us with the
603 novel use of 3D-printed charts. This newer type of visualization has gained a small amount
604 of traction in recent years as a method of producing tangible charts.

605 At this time, there are few, if any, studies that evaluate 3D-printed charts for the purpose
606 of displaying statistical information. This may in part be due to the cost of materials and
607 rendering times as compared to charts produced on a computer. A single chart can take up
608 to a day to print, limiting the ability to quickly produce visualizations. Given the nature of
609 3-dimensional data, viewing a 3D realization of statistical information in a 3D environment
610 is a realistic scenario that could increase the ability for information extraction.

611 3D-printed visualizations have shown mixed or promising results across many disciplines.

612 @katsioloudis2018 showed no evidence of a statistical difference in the method of 3D ren-
613 derings when tasked with drawing a cross-sectional of a dodecahedron. This demonstrated
614 that spatial awareness between computer-rendered and 3D-printed shapes is not largely dif-
615 ferent among engineering students. The use of 3D-printed maps for navigation for visually
616 impaired persons showed positive feedback by @holloway2019, increasing the accessibility of
617 navigation. In the clinical setting, 3D-printed anatomy structures were well accepted along
618 with VR-glasses and 3D computer renderings [@muff2022]. With the rise of 3D-printed vi-
619 sualizations in scientific fields, we turn our attention to their use for statistical graphics.

620 4.1.1 Literature Review

621 An identical dataset across multiple chart types does not ensure that the data is perceived in
622 the same way [@cleveland1984; @hofmann2012; @vanderplas2020]. One of the main factors
623 to this phenomena is with how data is encoded into the chart. These encodings include,
624 but are not limited to, placement along axes, lengths, areas, volumes, and color scales. For
625 example, bar charts and pie charts are two common visualizations that have long since been
626 the topic of debate [@eells1926; @croxton1927; @cleveland1984]. In the case of bar charts
627 and pie charts, encodings are represented by lengths and angles, respectively. Inherently,
628 the comparison of different chart types is a comparison of encodings due to the changes in
629 how data is being displayed.

630 @cleveland1984 noted that estimates involving numerical accuracy may decrease when in-
631 creasing dimensionality of the encoding, although this was not formally tested in their
632 experiments. The reasoning is possibly due to Stevens' power law, a mathematical formu-
633 lation of how magnitudes are perceived given different stimuli sources [@stevens1986]. The

634 general form of the law is $\psi(I) = kI^\alpha$, where I is the magnitude of a stimulus, $\psi(I)$ is
635 the perceived magnitude, k is a proportionality constant from the unit of the stimulus, and
636 α is the exponent from the type of stimuli used. Studies have estimated that lengths are
637 perceived without bias (i.e., $\alpha = 1$), but areas and volumes tend to have skewed perceptions
638 [cleveland1984]. This indicates that lower-dimensional charts might perform better when
639 readers are tasked with extracting numerical estimates from the chart.

640 There are mixed results in regard to the use of 3D charts, mostly attributing to the purpose
641 of third dimension. When the extra dimension does not convey meaningful information,
642 estimates of accuracy decrease and solution times increase as compared to equivalent 2D
643 charts [fisher1997; zacks1998; fischer2000]. The same increase in solution time is seen
644 when the third dimension is utilized for displaying data, but can sometimes produce better
645 error rates than 2D charts [barfield1989; kraus2020]. Additionally, when given the option
646 of 2D or 3D charts for extracting numerical information, the 2D charts showed increased
647 preference and confidence than their 3D counterparts [barfield1989; fisher1997]. It is
648 worth noting that all of the studies listed use renderings of 3D charts and not physical 3D
649 charts.

650 Formal studies involving true 3D charts are limited, and it is unclear if they follow the frame-
651 work of existing theories on data visualizations. Unlike paper and computer rendered charts,
652 constructing true 3D charts inherently contains many additional factors that could affect
653 perception, such as chart materials, natural lighting, interactivity, and viewing distance.
654 Some of these factors have already been shown to have an effect on computer renderings
655 [tarr2001; wang2022].

656 We hypothesize that 3D charts in 3D environments will produce better information extrac-

tion than their computer rendered counterparts. Specifically, we will compare 3D-printed charts to digital 2D and 3D renderings. We suspect that this difference will hold across multiple data sets and different magnitudes of stimuli. In this paper, we evaluate the accuracy of numerical estimations on true 3D charts by conducting a factorial experiment that assessed chart types and ratios of pairs of stimuli. We discuss the construction of the stimuli and how we closely matched the charts to compare 2D, 3D-digital, and 3D-printed renderings of heatmap data.

4.2 Methods

Our study is designed to evaluate and expand the literature on numerical estimation of 3D charts. In our study, our focus is on 2D and 3D heatmaps, which we carefully construct to ensure that differences are contributed to the dimensionality of the chart. All of our data and methods are publically available for open science and reproducibility at <https://github.com/TWiedRW/ch3-heat3d>. In this section, we discuss process of designing our experiment and participant recruitment.

4.2.1 Stimuli

We denote “stimuli” to represent the magnitude of our chosen values. In a 3D Cartesian space, the X- and Y- axes represent the coordinates of the stimuli, and the Z-axis represents the value for the stimuli. Each X and Y coordinate is represented by a square tile with a 1:1 aspect ratio. For a 2D space, the Z-axis is replaced by a color gradient scale. All stimuli and remaining randomly generated values range between 0 and 100 units. In this experiment, $X = 1, 2, \dots, 10$ and $Y = 1, 2, \dots, 10$.

678 The design of our experiment made use of the method of constant stimuli, where com-
 679 parisons between stimuli are with respect to a stimuli that remains the same magnitude
 680 [atkinson2016 chap. 3]. We set the constant stimuli at 50 units. For stimuli between
 681 50 and 100, we set the maximum stimuli value at 90 and equally partitioned the ratios of
 682 stimuli with the constant stimuli, $\frac{50}{90} = 0.556$ to $\frac{50}{50} = 1.0$, resulting in 4 varying stimuli
 683 values where 50 is the smallest value. The same ratios obtained with stimuli between 50
 684 and 90 were used to create 4 stimuli varying between 0 and 50, where 50 is the largest value.
 685 Additionally, we also included a stimuli pair where both values are 50, resulting in nine
 686 total pairs of stimuli. All stimuli values can be found in Figure 4.1.

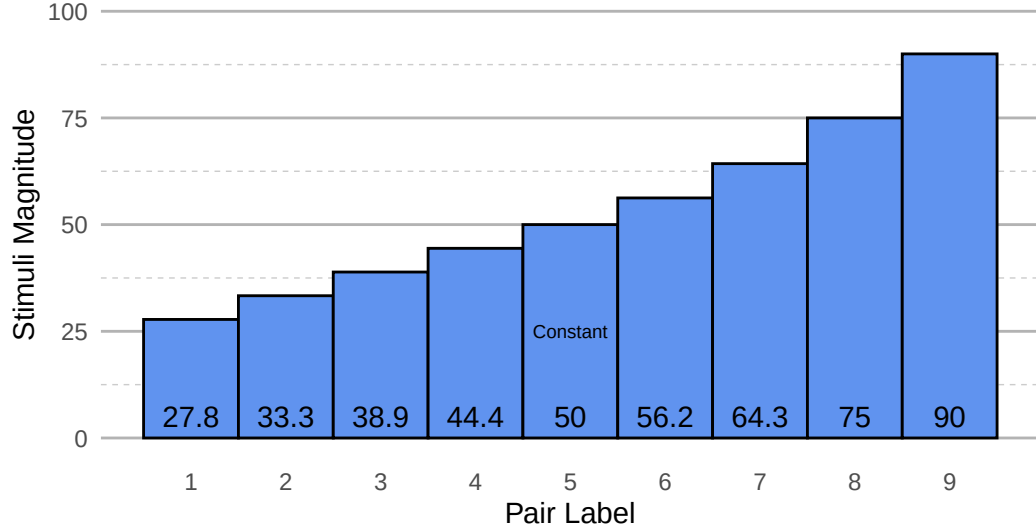


Figure 4.1: Values for stimuli in the heatmap experiment. All values are paired with the constant stimuli of 50 units, creating nine pairs of stimuli.

687 To generate non-stimuli random values, we used a mixture distribution of random uni-
 688 form noise and a mathematical function to populate our coordinate grid. The mathemat-
 689 ical functions are scaled between 0 and 100, $g(Z) = 100 \cdot \frac{Z - \min(Z)}{\max(Z) - \min(Z)}$. Two datasets
 690 were created for the experiment. The first dataset, called set 1, used the formula for
 691 the top half of sphere that is centered within our X and Y coordinate grid, $f_1(X, Y) =$
 692 $\sqrt{7^2 - (X - \bar{X})^2 - (Y - \bar{Y})^2}$, where $\bar{X}$ and $\bar{Y}$ are the averages of the X and Y coordinate

693 ranges. The second dataset is calculated similarly using the formula for the bottom half
694 of sphere, $f_2(X, Y) = \sqrt{7^2 - (X - \bar{X})^2 + (Y - \bar{Y})^2}$. Denoting Z as the random values,
695 $U(0, 100)$ as a random variable drawn from a continuous uniform distribution, and X, Y as
696 heatmap coordinates, our random heatmap data is calculated in Equation 4.1 with $c = 0.3$.
697 An example of varying c values is presented in Figure 4.2.

$$Z = c \cdot U(0, 100) + (1 - c) \cdot g(f_i(X, Y)) \quad (4.1)$$

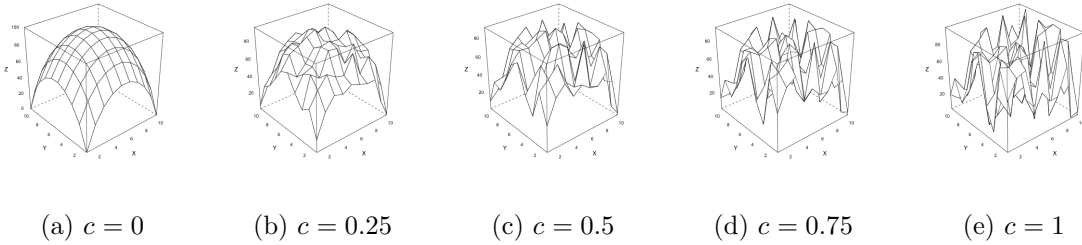


Figure 4.2: Mixture distribution of Equation 4.1 using the formula for the top half of a sphere. As c approaches 1, the distribution resembles uniform random noise.

698 The placement of stimuli values onto the randomly generated heatmap data was done via
699 simulation to try to make their placement look as natural as possible. Twenty random
700 heatmaps were generated. For each heatmap, the non-constant stimuli was placed onto
701 the coordinate such that the difference between the stimuli and randomly generated value
702 is minimized. The constant stimuli was then placed onto a coordinate with a Manhattan
703 distance of three or four that minimizes the difference between the constant stimuli and
704 randomly generated values, where the Manhattan distance is given by $|X_i - X_j| + |Y_i - Y_j|$
705 for stimuli i and j . To ensure that stimuli placement is evenly position across the heatmap,
706 the count of stimuli was computed separately across the X and Y axes. For example, in
707 Data Set 1, the X-axis has four stimuli in $X = 1$, three stimuli in $X = 2$, and so forth. Chi-

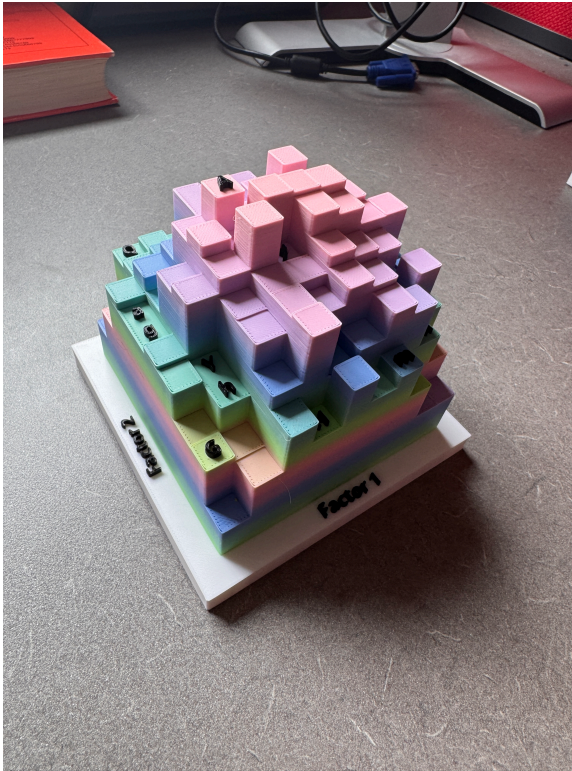
squared statistics were calculated for each axis and the heatmap with the smallest average Chi-squared statistic was selected as the final dataset. A visual inspection of this process showed that the stimuli were not clustered in any one area of the chart and that the stimuli look natural with respect to the random mixture distribution.

4.2.2 Charts

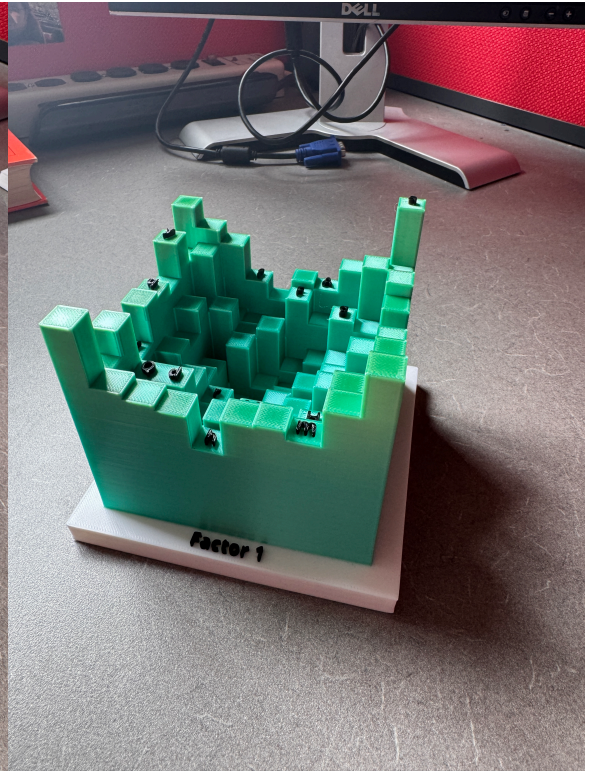
Three types of charts were considered for this study: 2D-digital (2dd), 3D-digital (3dd), and 3D-printed (3dp). We constructed these charts so that they are as similar as possible, but inherent difference between dimensionality led to artistic decisions that attempt to focus solely on the dimensionality of the charts. The process of creating the charts is discussed in this section.

The 3D-printed charts were rendered with OpenSCAD [kintelOpenSCADDocumentation2023]. To include plot text, a 120mm by 120mm by 10mm base was created with a solid color that was either white or black. Cells of the heatmap measured 10mm by 10mm, resulting in a heatmap that is 100mm by 100mm and is centered on the base. The upper bound of the height of heatmap values is 100mm, where 1-unit in the heatmap data is represented by 1mm of height on the heatmap. Once rendered, the heatmap was saved to 3D Manufacturing Format (3mf) and Standard Triangle Language (stl) files. A variety of solid and gradient filaments were used to print the output files from OpenSCAD. An example of the 3D-printed chart is shown in Figure 4.3.

To closely match the 3D-digital chart to the 3D-printed chart, multiple stl files were created for each colored component and combined with the RGL package [rgl]. The base was rendered with white smoke (#F5F5F5) to slightly contrast with the default white back-



(a) Solid filament of Data Set 1



(b) Solid filament of Data Set 2

Figure 4.3: 3D printed heatmaps.

730 ground color (#FFFFFF). Heatmap tiles were rendered with cyan (#74CCFF) and text
731 labels were rendered with black (#000000). Lighting was fixed at 45 degree angles at two
732 opposite corners of the chart. The end result was a near perfect replica of the 3D-printed
733 charts, with the exception of different heatmap tile colors, where an example is given in
734 Figure 4.4.

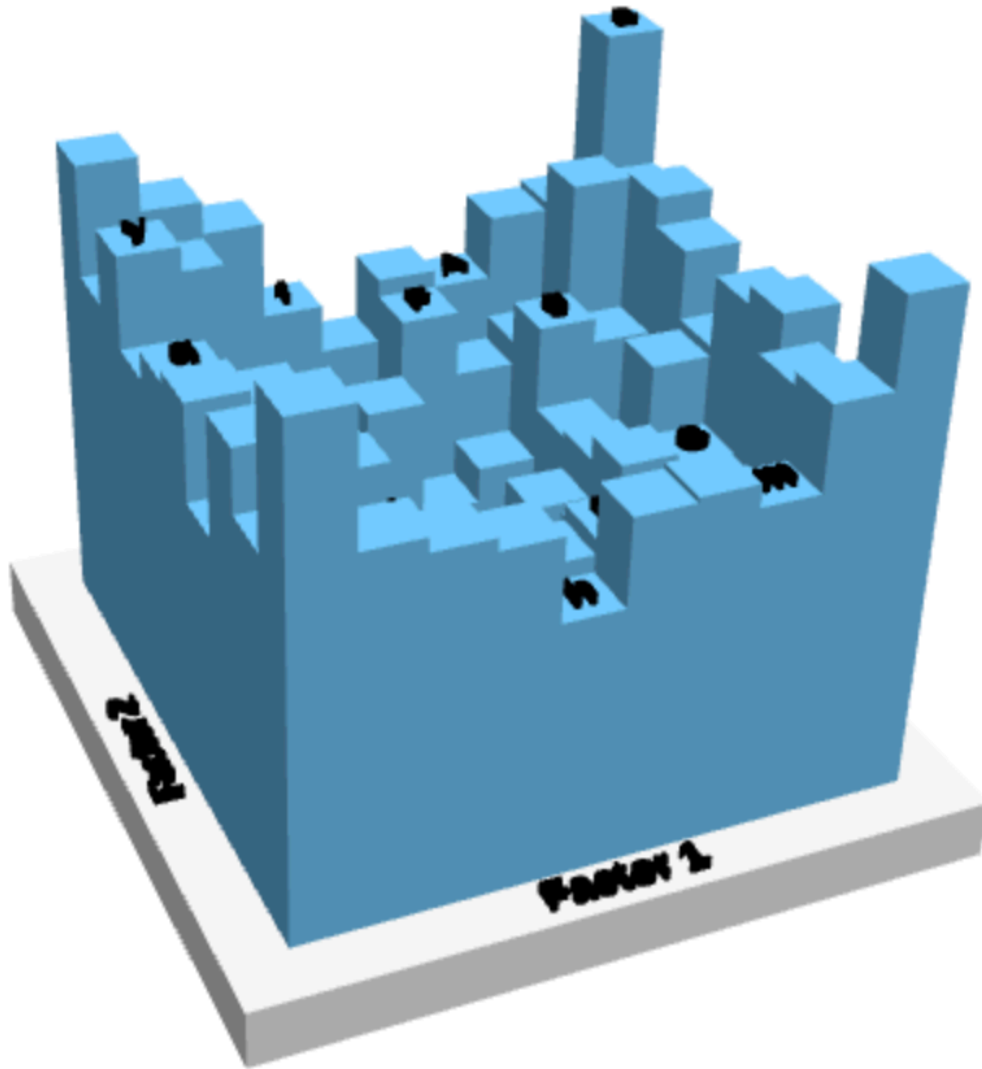


Figure 4.4: RGL rendering of Data Set 2 for 3dd charts.

735 Unlike the 3D charts, the 2D charts needed a different encoding to convey heatmap values.

736 The 2D heatmaps were created with `ggplot2` [ggplot2] using `geom_tile()`. Fill colors for
737 the cells use a color gradient from Blue Zodiac (#0C2841) to Malibu (#66D9FF), which
738 were selected from a color picker using shadows on our initial 3D-printed charts. The color
739 interpolation was performed with the `scale_fill_gradient()` function from the `ggplot2`
740 package.



Figure 4.5: Color palette for 2D-digital charts. The colors are interpolated from #0C2841 to #66D9FF, which are the the colors of the lighting conditions for a 3D-printed chart created with cyan filament.

741 4.2.3 Subject Recruitment

742 Participants were recruited from all STAT 218: *Introduction to Statistics* sections at Univer-
743 sity of Nebraska-Lincoln from June 2025 to Spring 2026. The experiment was incorporated
744 into the curriculum as a project that allowed students to get hands-on experience with
745 statistics. Throughout the project, students were also given a series of reflections to eval-
746 uate their perspectives on statistics. The first reflection had gathered students' thoughts
747 on the scientific process before participating in the experiment. After the experiment par-
748 ticipation, students reflected on what they thought the experiment was measuring. Lastly,
749 students were given a written article and video presentation of the experiment and asked
750 how their view of the experiment changed from their initial perspectives of the experiment's
751 purpose.

752 To assess whether students completed the experiment or not, the experiment produced
753 a completion code for students to submit to Canvas. The completion code was created
754 by concatenating three words with dashes (e.g., palm-raising-creatively). Words for the

755 completion code came from the movie transcripts of the Star Wars prequel movies (Episodes
756 1, 2, and 3) as presented by movies.fandom.com. The movie transcripts were tokenism and
757 filtered to remove colorful language¹, stop words², and fictional words³. Instructors were
758 provided an R script to check if their students completed the experiment, but the completion
759 codes were not saved with any identifying information that could link experiment responses
760 to student reflection responses.

761 4.2.4 Experimental Design

762 Our experiment was designed with a 3 x 2 x 9 treatment structure. Media type is our main
763 interest, with 2D-digital, 3D-digital, and 3D-printed charts. To ensure that results are not
764 confounded with datasets, we used two datasets to create the heatmaps. A total of 9 stimuli
765 pairs were placed into each dataset. The order of treatments was given so that media and
766 dataset combinations were grouped together randomly in the sequence and stimuli pairs
767 were randomized within the groupings.

768 Due to practical constraints, stimuli pairs were incompletely blocked. A full factorial design
769 would result in 54 trials per participant, which could lead to a decrease in quality responses
770 [Herzog1981]. Therefore, we selected four out of the nine possible stimuli pairs to create
771 incomplete blocks. This resulted in 18 blocks for a balanced incomplete block design. Within
772 each block, media type is fully crossed with dataset. Using the incomplete block structure,
773 participants completed a total of 24 trials, which is more practical than the full factorial
774 design. Blocks were chosen for each participant by using probabilities that are proportional
775 to the inverse counts of used blocks, meaning

¹<https://www.cs.cmu.edu/~biglou/resources/bad-words.txt>

²tm package [tm]

³@web2_dict

776 We measured three responses – two questions for each trial and one question for each media
777 by dataset combination. For each trial, participants are initially asked which stimuli in
778 a pair is larger or if the stimuli are the same value. Next, they were asked to estimate
779 the magnitude of the smaller stimuli if the larger stimuli represents 100 units, which is a
780 subtractive process [a@veit]. After each grouping of media and dataset, participants were
781 presented with a modal dialog window asking them to rate their confidence on a 5-point
782 Likert scale for their answers of the previous group of charts.

783 4.2.4.1 Shiny Application

784 A Shiny application [a@shiny] was developed to administer the experiment. The application
785 consisted of five sections: informed consent, demographics, practice, experiment, and wrap-
786 up. The entire application is designed to be completed in approximately 30 minutes.

787 The Shiny application started with the informed consent screen, allowing participants to
788 select if they are a STAT 218 Student and if they agree to the data collection. Participants
789 had to select a data collection option to continue. After submitting their data collection
790 response, a completion code was generated and saved on the last page of the application.
791 A copy of the informed consent is available in our GitHub repository.

792 If participants agreed to have their data collected, they were presented with the demo-
793 graphics section. This section asked participants to use drop down menus to specify their
794 age, gender identity, highest education level, and a question about how their participation
795 in the experiment is graded. The last question was a text box and asked participants to
796 specify their favorite movie and/or actor. Demographic information was combined with the
797 application start time and completion code to create a hash for an anonymized participant

798 identifier using the `rlang` package [`@rlang`].

799 Once a participant completed the demographics page or selected “No” to the data collection
800 question, participants were given a practice page. A modal dialog window was initially
801 shown with instructions, and users could display this window again at any point during
802 the practice. The practice consisted of four questions: two trials from 2dd charts and two
803 trials from 3dd charts. Each practice trial showed the correct solution after the participant
804 submits their trial. After all practice trials were completed, another modal dialog window
805 was displayed to ask participants if they have access to the 3D-printed charts. This was
806 necessary since the experiment was given to potentially online sections of STAT 218 who
807 did not have access to the physical charts.

808 The experiment page was presented to participants after completion of the practice trials
809 Figure 4.6. Each page contained two questions – one question for identifying which stimuli
810 is larger and another question for estimating the value of the smaller stimuli if the larger
811 stimuli is 100 units. If participants selected that the stimuli are the same value, then the
812 slider was automatically placed at 100 units. Each trial has the slider initially placed at
813 50 units. An option was provided for displaying the coordinates of the stimuli pair on a
814 2D heatmap, but was left unchecked by default. For 3dd charts, the number of interactive
815 clicks was also recorded. A trial could only be submitted if the first question is answered
816 and if the slider was moved at least once.

817 After completing all trials, participants were given the completion code and informed to
818 copy the code to Canvas since they will not have access to the code after closing out of the
819 experiment.

Trial 1 of 24

Group 1 of 6

Which of the following values is larger?

☐ F

☐ f

☐ The two values are the same.

If the larger value you selected above represents 100 units, how many units does the smaller value represent?

0 50 100

Submit

☐ Select this checkbox if you need help identifying the two values on the chart.

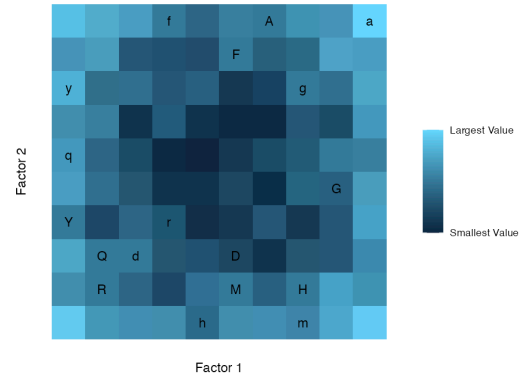


Figure 4.6: Graphics experiment page. Each trial contained two questions about a pair of stimuli, and a check box that displays the location of the stimuli if needed.

820 4.2.4.2 Method of Analysis

821 The design of our experiment is a balanced incomplete split-plot with replication, where
 822 media type and dataset is the whole plot and stimuli pair is the split plot. A similar design
 823 was presented by @mandal2020 for a single replicate. For our experiment, the ANOVA
 824 table is in Table 4.1.

Table 4.1: ANOVA table for analysis of a single replicate of the balanced incomplete split-plot.

Source	Degrees of Freedom
Block (B)	18-1
Media type (M)	3-1
Data Set (D)	2-1
Whole-plot error = $B \times M \times D$	$(18-1)(3-1)(2-1)$
Stimuli Pair (P)	9-1

Source	Degrees of Freedom
$P \times M$	$(9-1)(3-1)$
$P \times D$	$(9-1)(2-1)$
$P \times M \times D$	$(9-1)(3-1)(2-1)$
Split-plot error = $B \times M \times D \times P$	$(18-1)(3-1)(2-1)(9-1)$

825 The first question for each trial was to indicate which value within a stimuli pair was
826 larger, or if they were the same value. Since a participant could either answer this question
827 correctly or incorrectly, a Binomial generalized linear mixed model [atou] was fit to the
828 responses.

829 To measure numerical accuracy of participant responses for the second question in each trial,
830 we used the absolute value of the difference between the participant estimate of the smaller
831 stimuli value and its true value. By using the absolute value of the error, we measure how far
832 off a participant's estimate was rather than the direction of error. For numerical accuracy,
833 we fit a linear mixed model. If the responses do not fit well to a Gaussian distribution, other
834 distributions, such as the log-normal or gamma distributions, will be used for a generalized
835 linear mixed model.

$$Y = |\text{Guess} - \text{Actual}|$$

836 In addition to linear mixed models, we used the bootstrap distribution of mean errors as
837 described by @cleveland1984 to provide empirical claims for the effect of dimensionality. A
838 common issue in experiments involving perception is that random noise often covers the

839 effects of interest, so the empirical claims may provide additional insight into the effect of
840 chart dimensionality (XXX).

841 After each media type and data set grouping, participants were given a question to rate
842 their confidence on their answers. Here, we use descriptive statistics and visualizations to
843 assess confidence across our treatment factors.

844 **4.3 Limitations**

845 Our study was designed to evaluate the accuracy of numerical estimation of stimuli pairs
846 across three types of charts. However, numerical accuracy is not the only deciding factor in
847 what makes a good chart design. Other factors include comprehension, recall, and overall
848 preference [fisher1997; vanderplas2020]. Given the lack of current research into 3D-
849 printed statistical graphics, we hope to provide some baseline insights into their use as a
850 statistical tool.

851 We did not account for potential visual illusions in the construction of the heatmap data.
852 A phenomenon seen in statistical graphics studies is that surrounding values may attribute
853 to participant estimates [zacks1998; vanderplas2015]. In our study, the placement of
854 stimuli was so that the stimuli looked naturally positioned within the structure of the data,
855 resembling patterns typically seen in spatial data. However, consistency of neighboring
856 values was not accounted for in this experiment and would be worth having a follow up
857 study to measure the effect of surrounding values on 3D-printed charts.

858 Lastly, we wanted to construct heatmaps so that they closely resembled each other across
859 dimensionality. The 3dd charts are identical to the 3dp charts, with the exception of

860 heatmap colors. Here, the issue is that 3D heatmaps are not typically constructed by
861 compiling stl files, but rather using packages such like `lattice` package [lattice] to display
862 the data. Due to parameter settings and code limitations, it may not be possible to get
863 typical 3D renderings of charts to appear visually identical to 3D-printed counterparts.

864 **4.4 Conclusion**

865 Our study provides insight into the novel use of 3D-printed statistical graphics. We hope
866 that these charts may be useful for educational purposes, accessibility of visually impaired
867 persons, or recreational use of statistical communication.